

Quiz: Indexes (2)

Name:

Quiz Rules:

1. you MAY use an printed or hand written material
2. you MAY NOT use a computer or other electronic device

All of the problems assume the following schema.

```
CREATE TABLE users (  
    id_users BIGINT UNIQUE NOT NULL,  
    created_at TIMESTAMPTZ,  
    username TEXT NOT NULL  
);  
  
CREATE TABLE tweets (  
    id_tweets BIGINT PRIMARY KEY,  
    id_users BIGINT REFERENCES users(id_users),  
    in_reply_to_user_id BIGINT REFERENCES users(id_users),  
    created_at TIMESTAMPTZ,  
    text TEXT  
);  
  
CREATE TABLE tweet_tags (  
    id_tweets BIGINT REFERENCES tweets(id_tweets),  
    tag TEXT,  
    UNIQUE(id_tweets, tag)  
);
```

1. Recall that certain constraints create indexes on the appropriate columns. List the equivalent `CREATE INDEX` commands that are run by these constraints.

Note: When solving all subsequent problems, assume only the indexes above have been created. For example, if problem 3 asks you to create an index, you should not assume that index has been created when working on problem 4.

2. List the scan methods applicable for the following SQL query.

```
SELECT count(*)  
FROM tweet_tags  
WHERE tag=:tag  
      AND id_tweets=:id_tweets
```

3. Create index(es) so that the following query can use an index only scan.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT count(*)  
FROM tweet_tags  
WHERE :tag = reverse(tag);
```

4. Create index(es) so that the following query can use an index only scan.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT count(*)  
FROM tweet_tags  
WHERE reverse(:tag) = tag;
```

5. Create index(es) so that the following query can use an index only scan, avoid an explicit sort, and take advantage of the LIMIT clause for faster processing.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT username
FROM users
WHERE id_users >= :start_id
      AND id_users < :end_id
ORDER BY id_users ASC
LIMIT 10;
```

6. Create index(es) so that the following query runs as fast as possible.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT text
FROM tweets
JOIN users USING (id_users)
WHERE username = :username
LIMIT 10;
```

7. Create index(es) so that the following query will run as efficiently as possible. Assume that you do not have access to RUM indexes.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT count(*)
FROM tweets
WHERE to_tsvector('english', text)
      @@ to_tsquery('english', :lexeme);
```

8. Create index(es) so that the following query will run as efficiently as possible. Assume that you do not have access to RUM indexes.

Do not create any unneeded indexes; if no new indexes are needed, say so.

```
SELECT count(*)
FROM tweets
WHERE to_tsvector('english', text)
      @@ to_tsquery('english', :lexeme)
      AND created_at > :created_at;
```